Chapter 1: BMW 730 steering dynamics

This report studies the two-state linear lateral model of the BMW 730 around a selected operating speed. The states are the lateral velocity v_y and the yaw rate omega, while the steering angle delta is used as the control input. The analysis is organised around internal stability, reachability, observability, transfer-function properties and simple active steering controller design.

1.1 State-space representation

$$x = [\ v_y \quad omega\ ]^T, \quad x_dot = A\,x + B\,delta$$

$$A = [[-alpha/V,\ -(V - beta/(mV))],\ [beta/(JV),\ -gamma/V]], \quad B = [Cf/m,\ Cf\,Lf/J]^T$$

The output matrix is selected according to the measured quantity. For yaw-rate measurements C_omega = [0 1], while for lateral speed C_vy = [1 0]. The parameter values used in the calculations are m = 1864 kg, J = 3654 kg m^2, Cf = 101600 N/rad, Cr = 213800 N/rad, Lf = 1.51 m and Lr = 1.32 m.

Parameter	Value
m	1864 kg
J	3654 kg m^2
Cf	101600 N/rad
Cr	213800 N/rad
Lf	1.51 m
Lr	1.32 m

1.2 Eigenvalues and open-loop stability

The characteristic polynomial of A is a second-order polynomial whose coefficients depend on the vehicle speed V. For forward motion the eigenvalues are in the left half-plane; at low speed they are real and at higher speed they become a complex conjugate pair. In reverse motion the signs change and the model becomes unstable.

$$p(s) = det(sI - A) = s^2 - (a11 + a22)s + (a11\,a22 - a12\,a21)$$

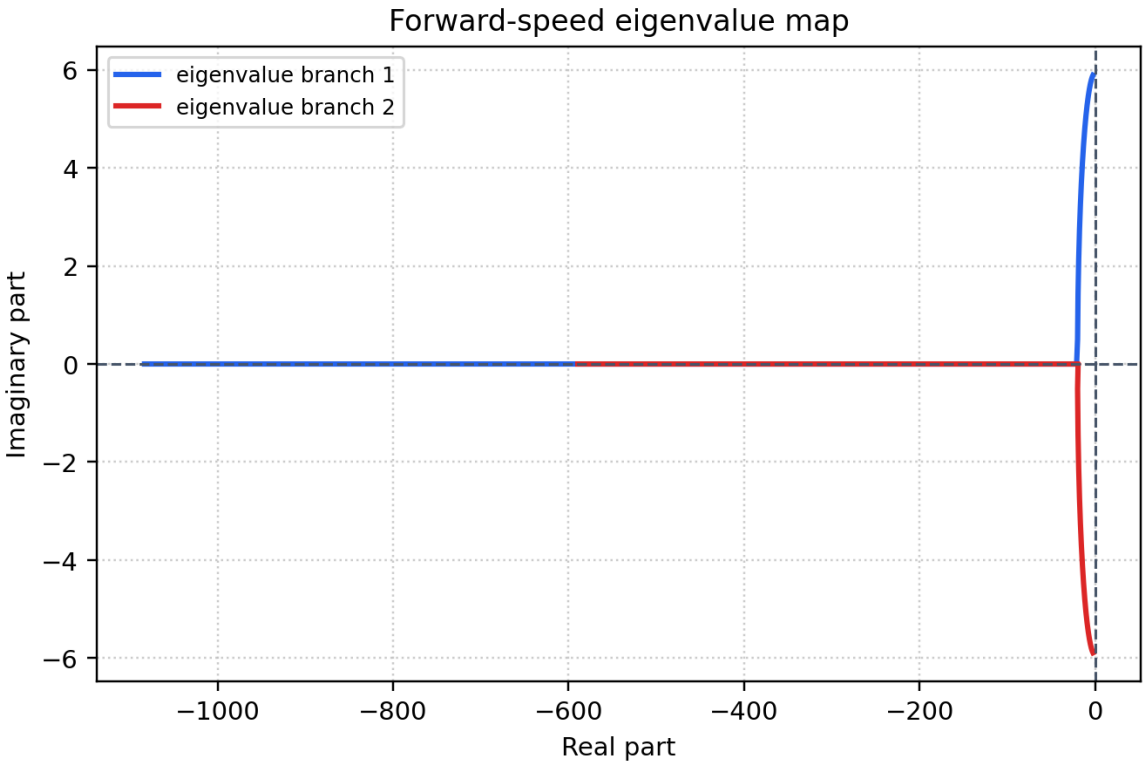


Figure 1. Forward-speed eigenvalue migration.

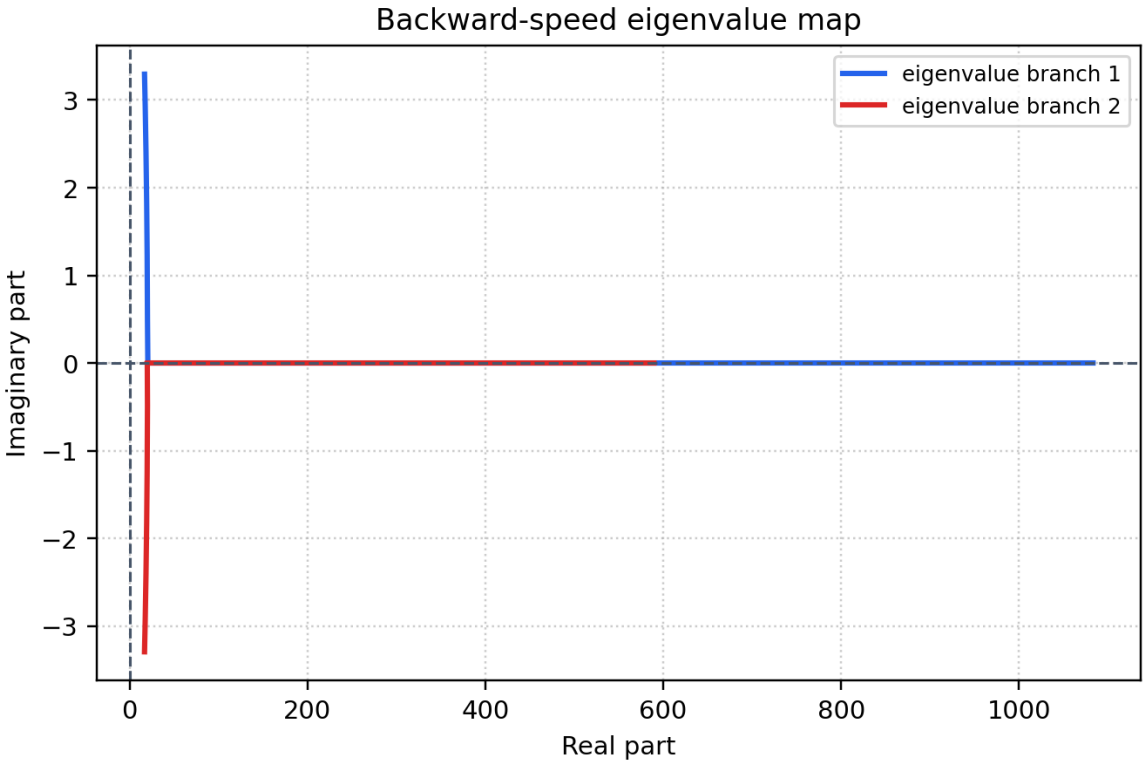


Figure 2. Backward-speed eigenvalue migration.

1.3 Reachability and observability

Reachability is evaluated using R = [B AB]. Observability from yaw rate is evaluated using O = [C; CA]. The determinant plots show where rank loss can occur. The model stays observable for the examined speed interval, while the reachability determinant crosses zero near low speed; therefore the zero-speed singularity and very low speed range must be treated carefully.

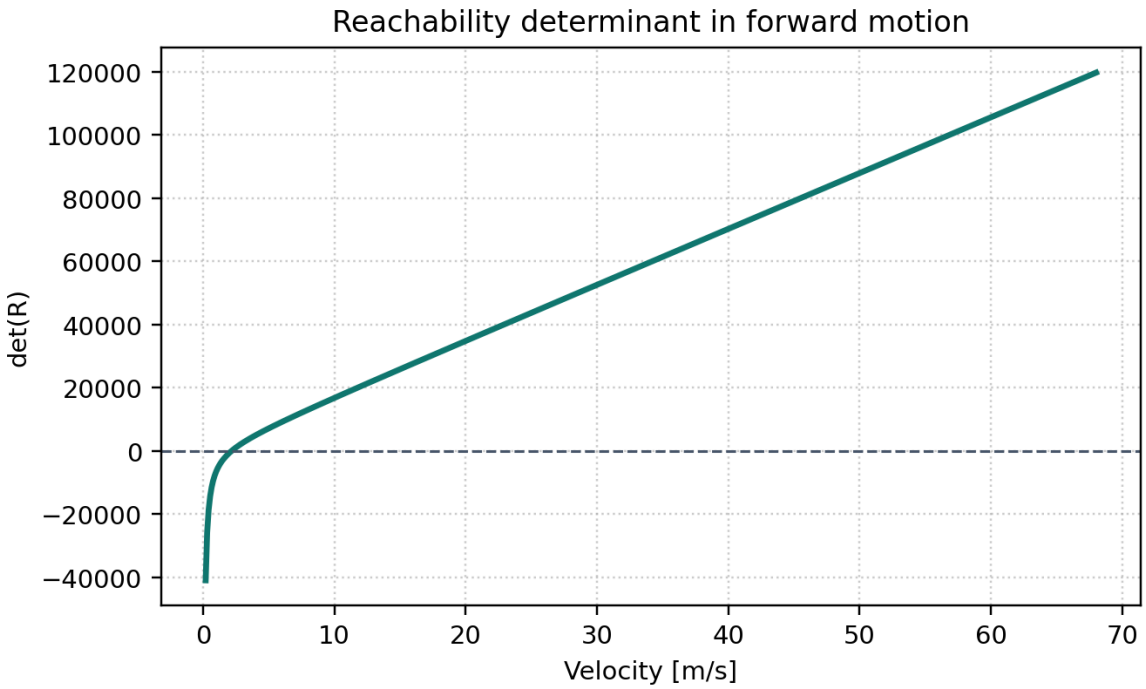


Figure 3. Determinant of the reachability matrix for forward speed.

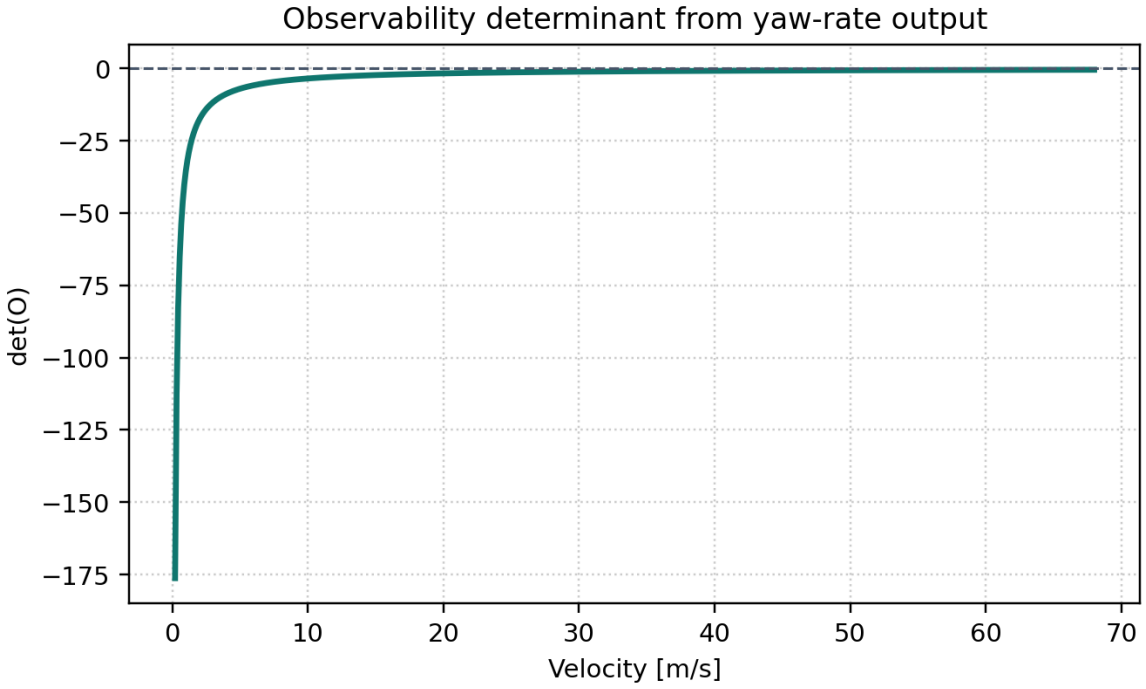


Figure 4. Determinant of the yaw-rate observability matrix for forward speed.

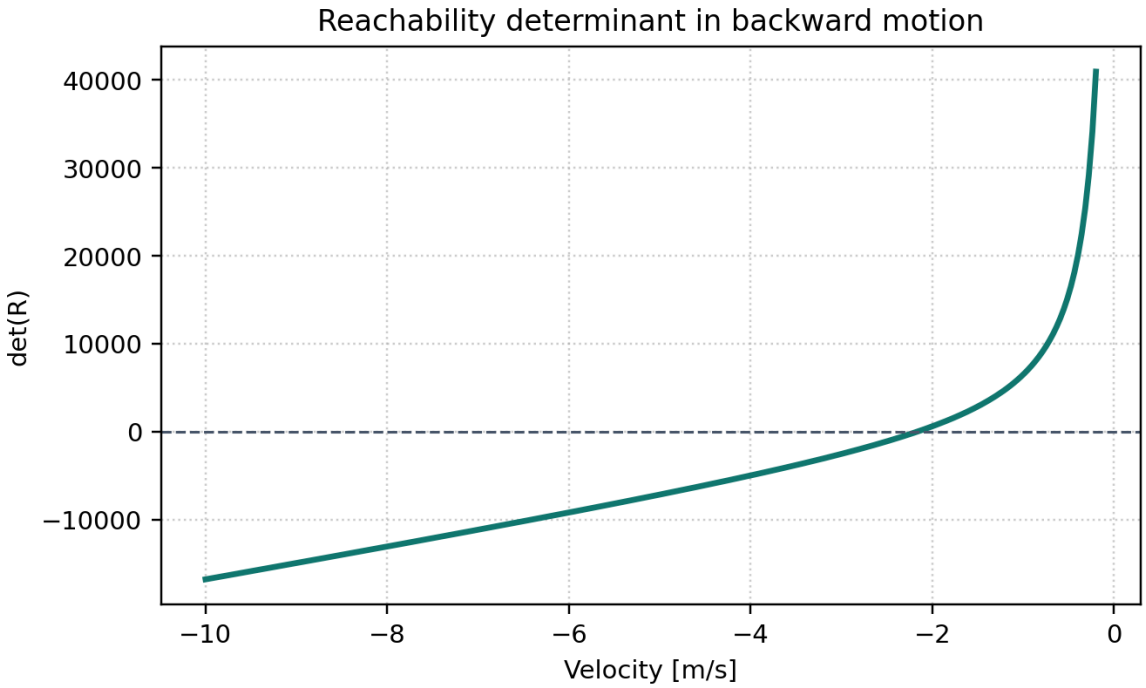


Figure 5. Determinant of the reachability matrix for backward speed.

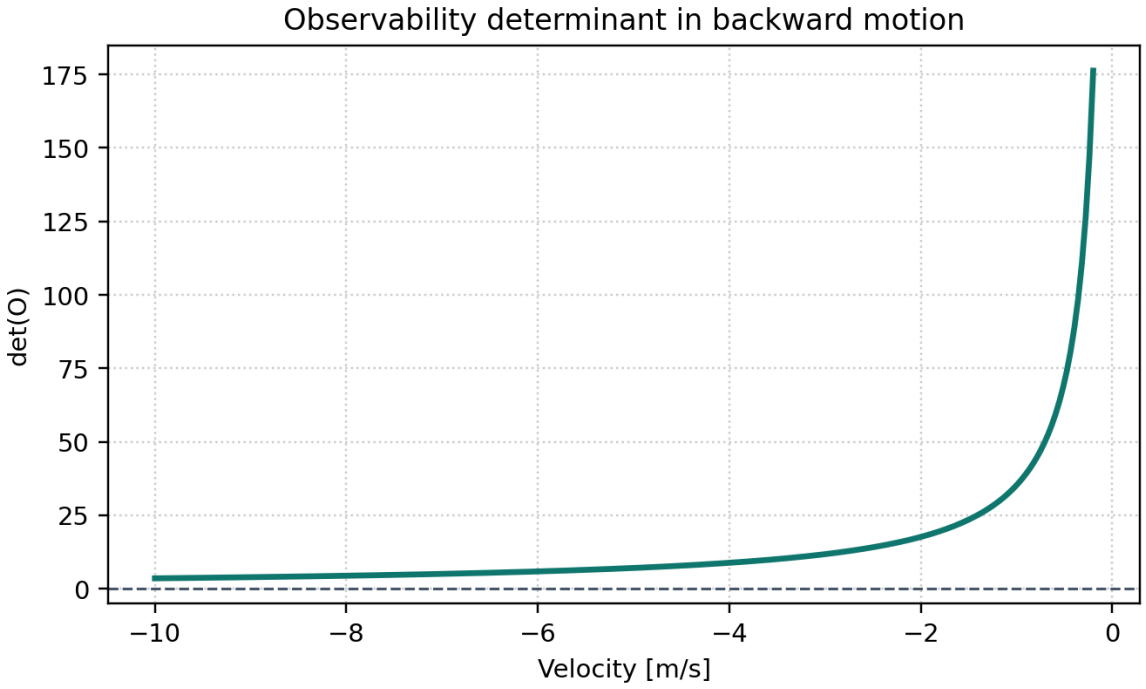


Figure 6. Determinant of the yaw-rate observability matrix for backward speed.

1.4 Transfer functions and minimum-phase properties

Two transfer functions are considered: W1(s) from steering angle to yaw rate and W2(s) from steering angle to lateral velocity. Both transfer functions share the same denominator, because they are generated from the same state matrix. Their zeros are different, so the minimum-phase conclusion depends on the selected output and vehicle direction.

$$W1(s) = (b2\ s + a21\ b1 - a11\ b2) / (s^2 - (a11+a22)s + a11a22-a12a21)$$

$$W2(s) = (b1\ s + a12\ b2 - a22\ b1) / (s^2 - (a11+a22)s + a11a22-a12a21)$$

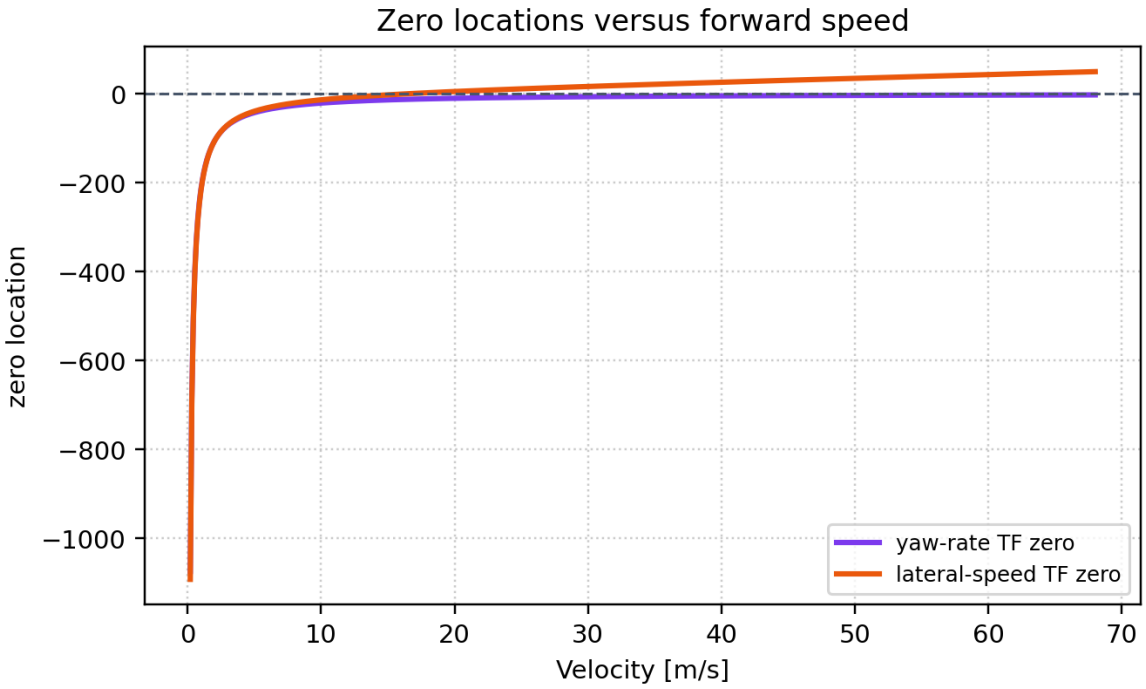


Figure 7. Zero locations of yaw-rate and lateral-speed transfer functions in forward motion.

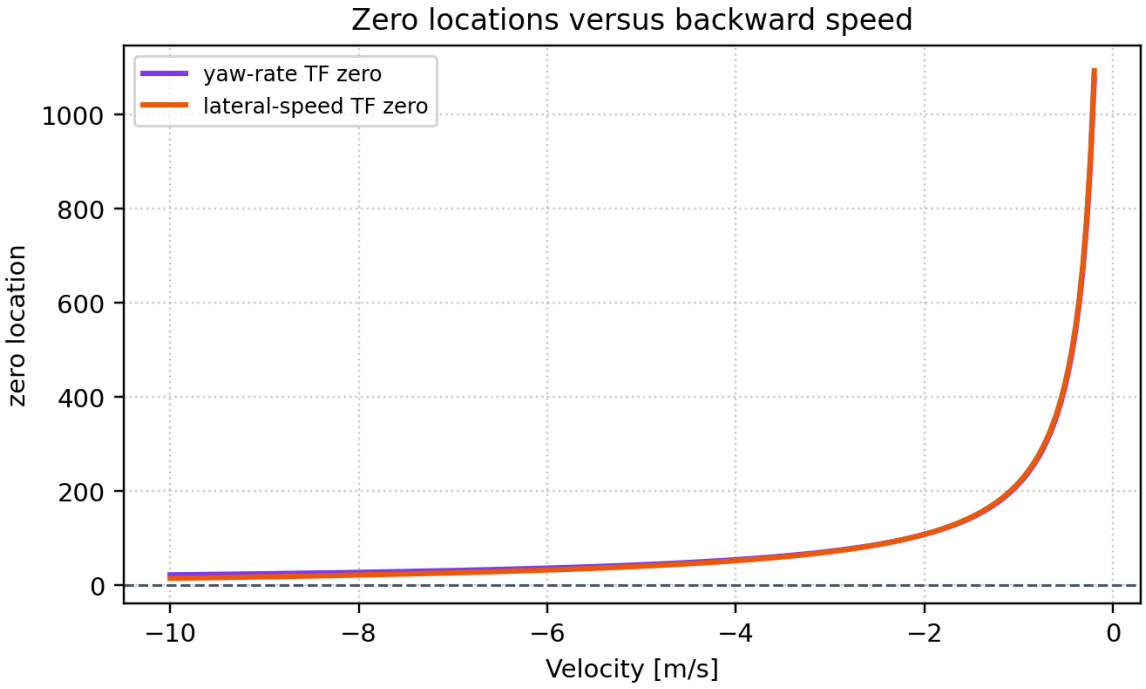


Figure 8. Zero locations of yaw-rate and lateral-speed transfer functions in backward motion.

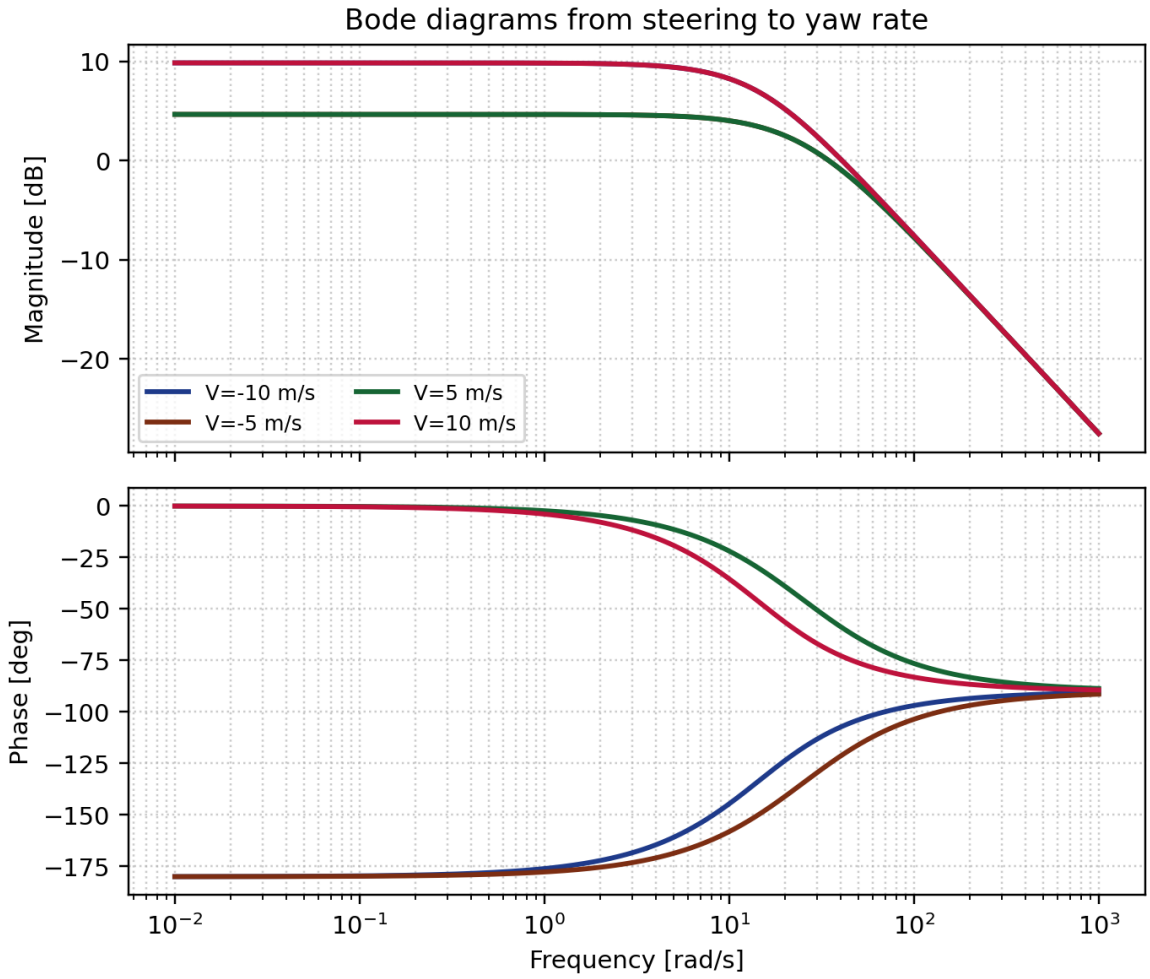


Figure 9. Bode diagrams for steering-to-yaw-rate transfer function.

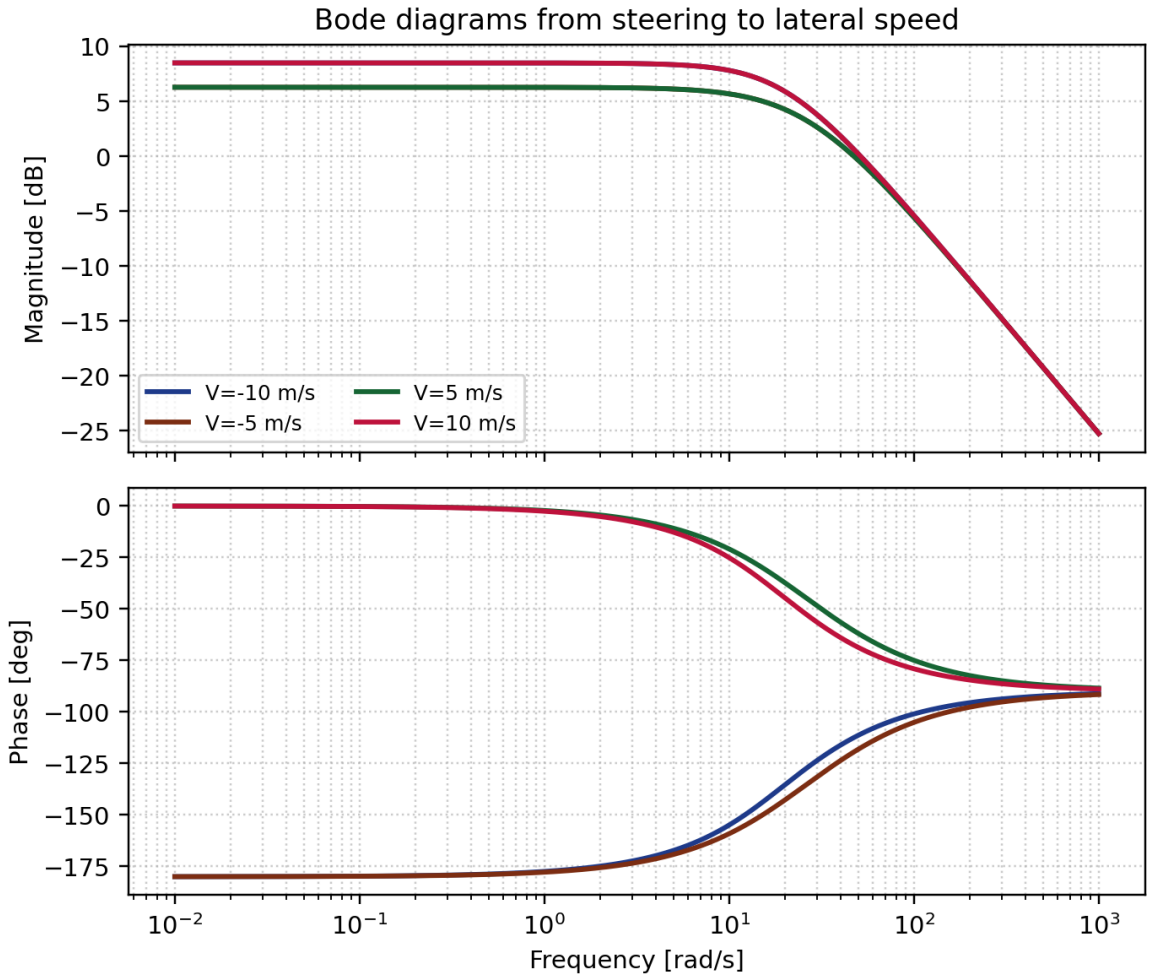


Figure 10. Bode diagrams for steering-to-lateral-speed transfer function.

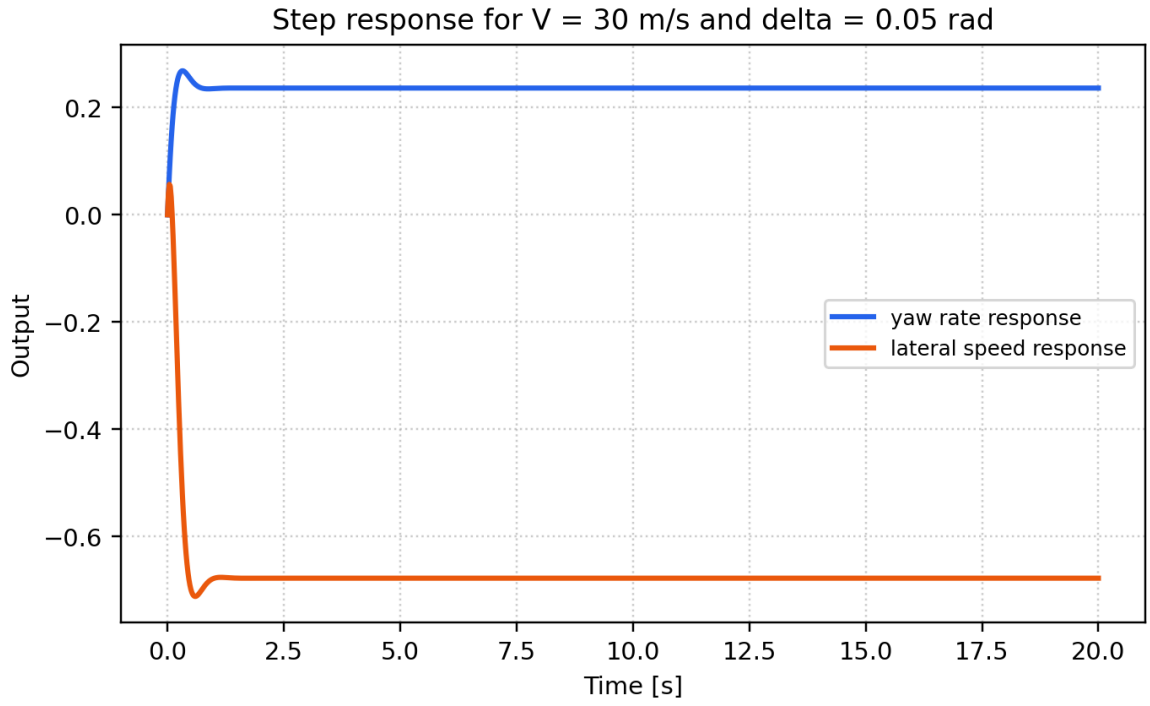


Figure 11. Step response for V = 30 m/s and delta = 0.05 rad.

1.5 Steering behaviour and damping

The static lateral-speed gain is used as an indicator for oversteering, neutral steering and understeering. The damping ratio decreases when the vehicle speed increases, so the model becomes more oscillatory at higher velocity even though the forward-speed open-loop poles remain stable.

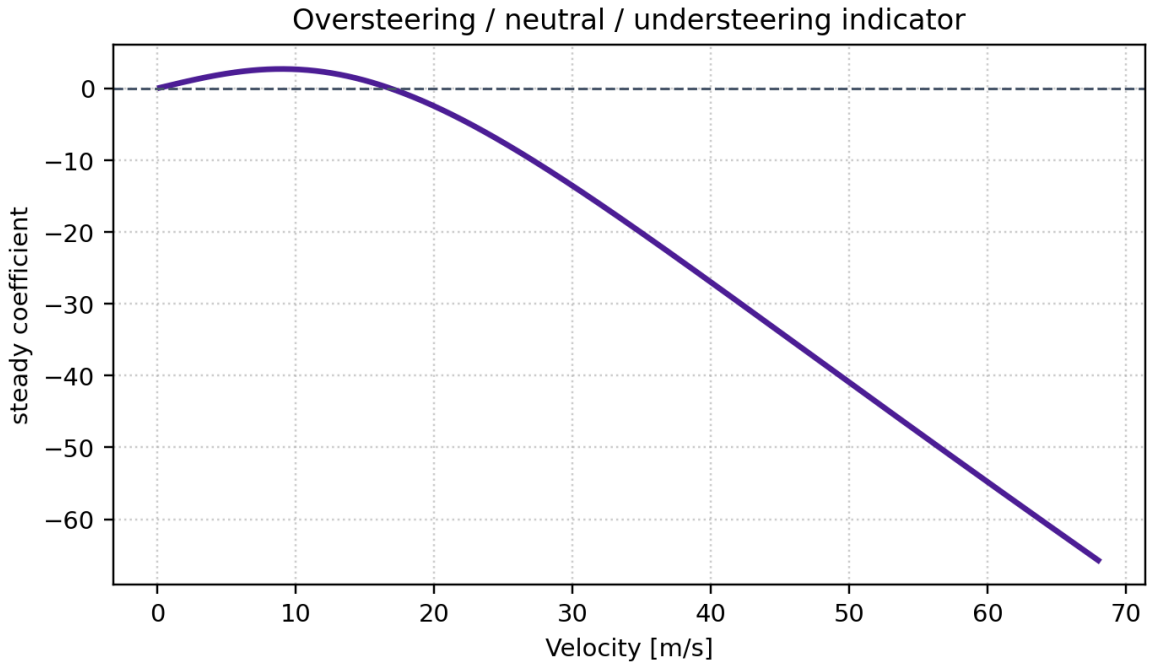


Figure 12. Static steering indicator versus speed.

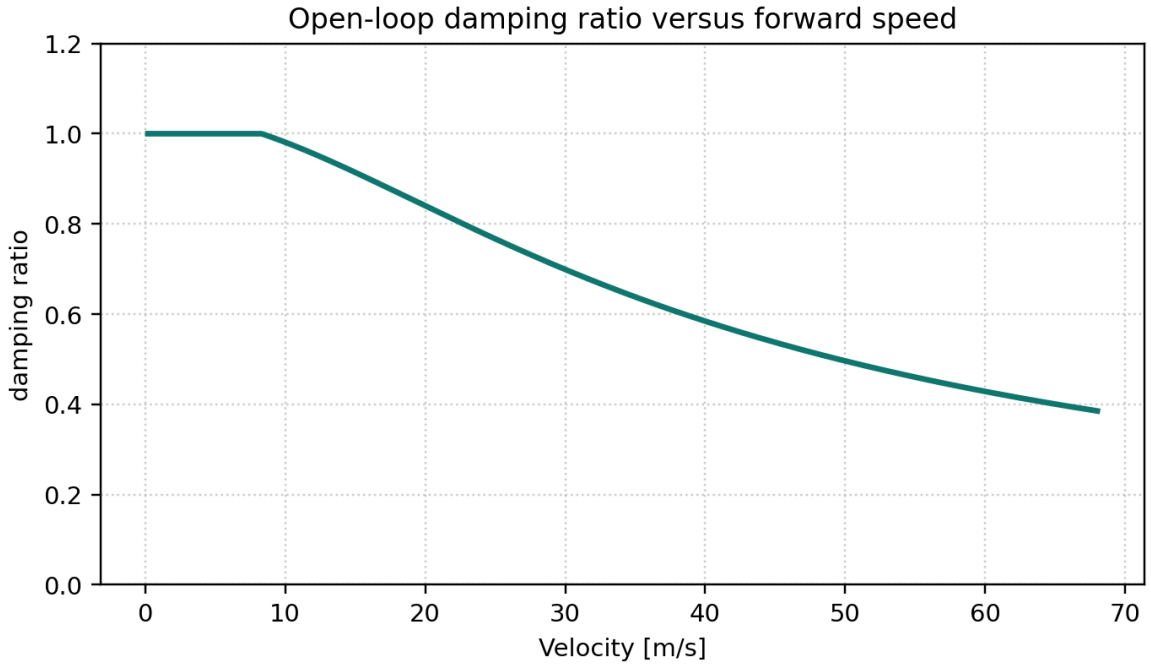


Figure 13. Damping ratio versus forward speed.

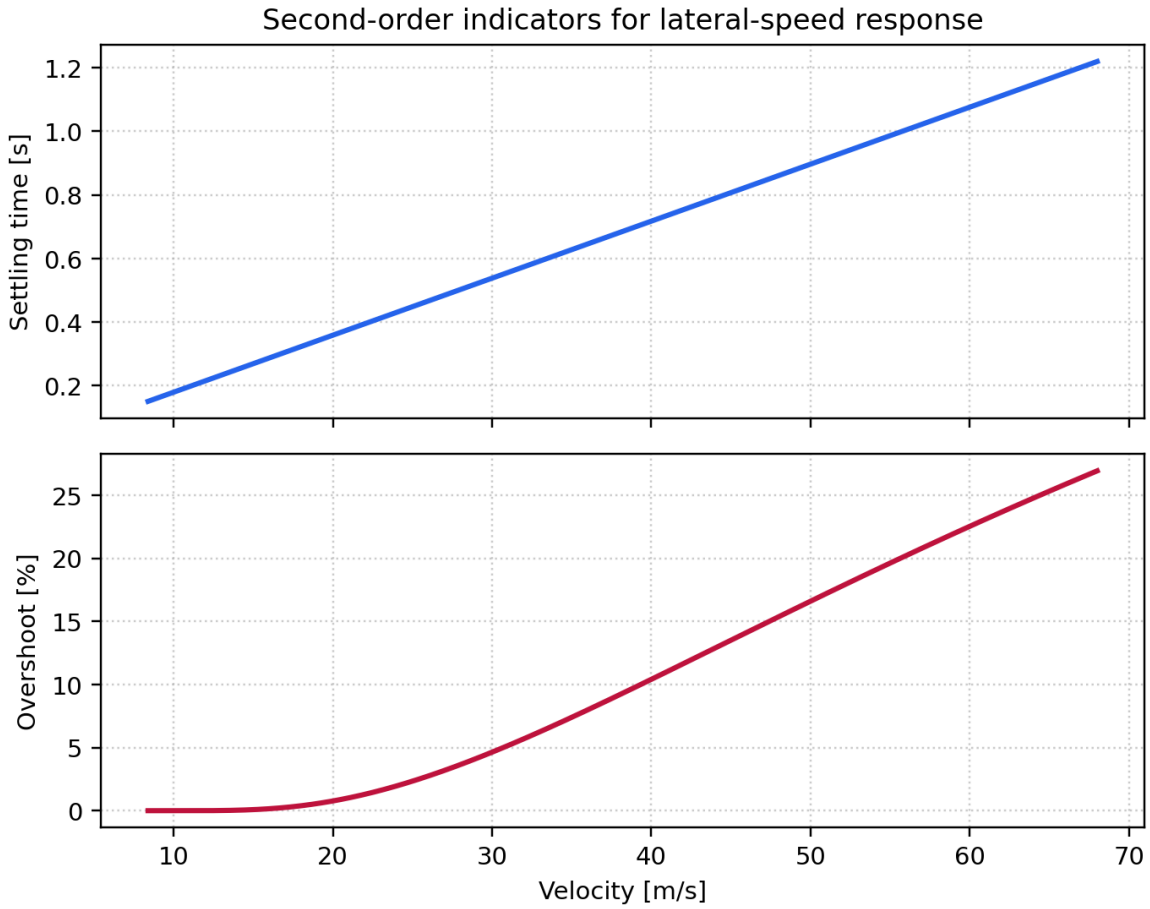


Figure 14. Settling-time and overshoot indicators.

1.6 Effect of road friction coefficient mu

When the tire-road friction coefficient is reduced, the lateral force coefficients effectively decrease. This shifts the eigenvalues and reduces the available damping. The model therefore predicts a larger need for careful feedback control on wet or icy roads.

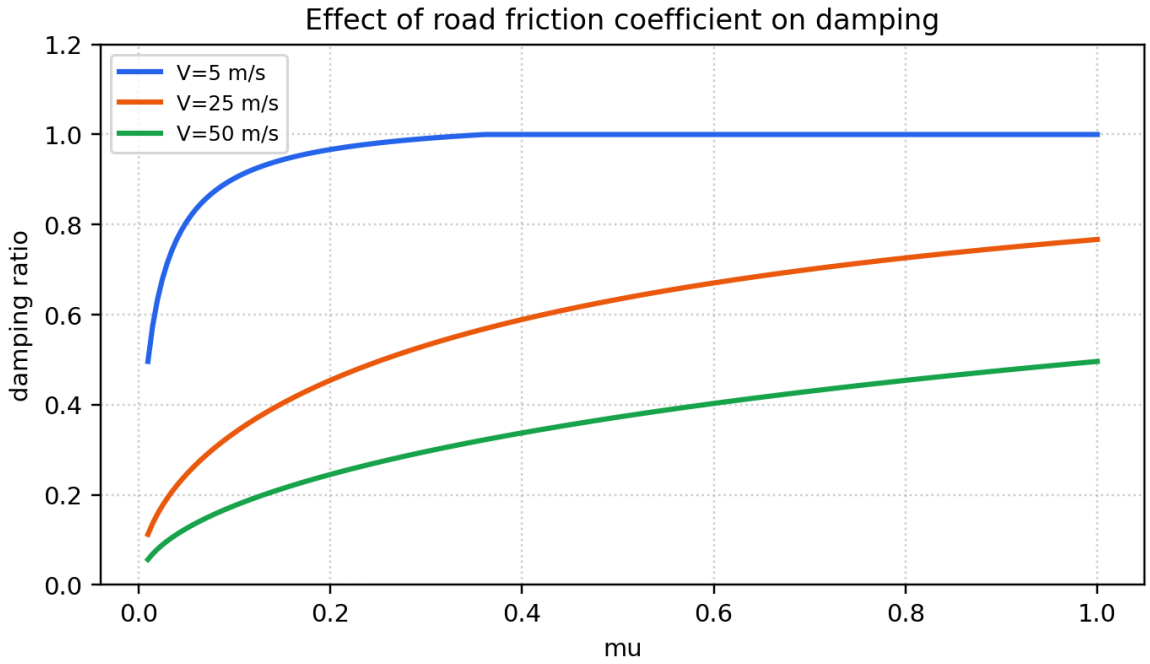


Figure 15. Damping ratio as a function of mu for three velocities.

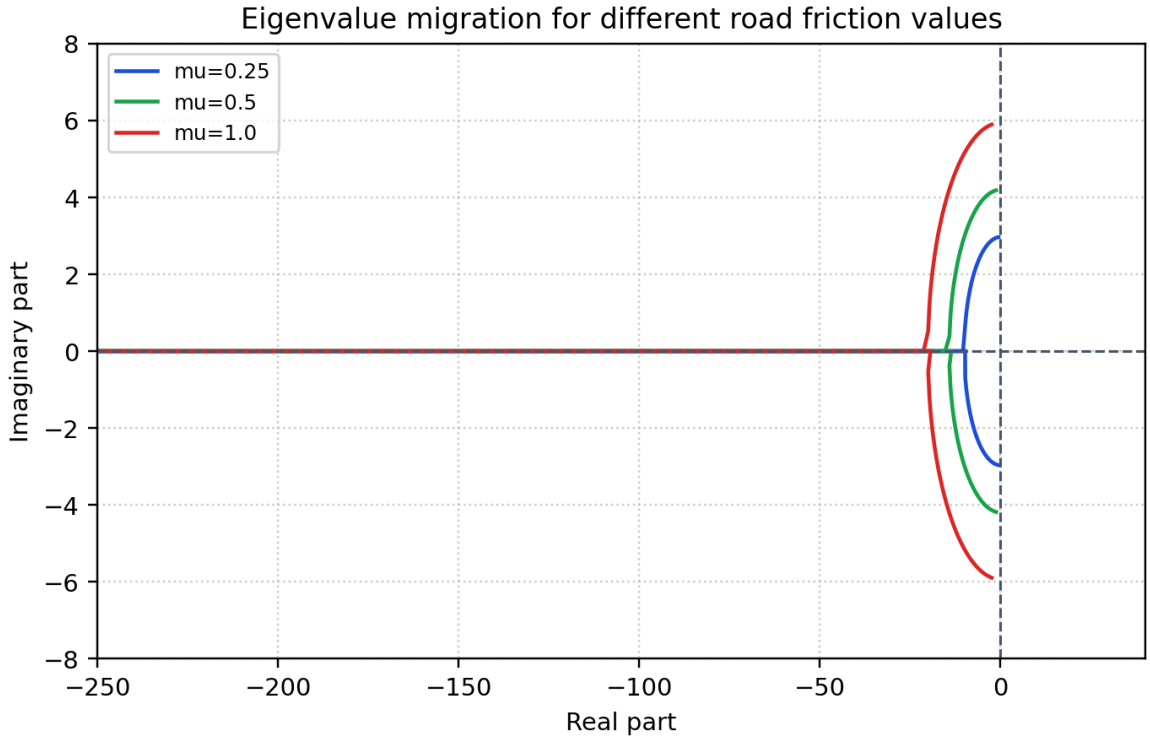


Figure 16. Eigenvalue migration for mu = 0.25, 0.5 and 1.

1.7 Feedback-control design for low-friction conditions

For a very low-friction case, mu = 0.001 and V = 10 m/s, the open-loop damping is poor. Root-locus based compensators are used to move the dominant closed-loop poles further into the left half-plane. For the yaw-rate output an additional zero is used with an integrator, while the lateral-speed controller also includes a low-frequency pole to improve the final pole distribution.

$$C_{omega}(s) = K\,(s + 0.1)\,/\,s$$

$$C_{vy}(s) = K\,(s - 3)\,/\,(s(s + 0.03))$$

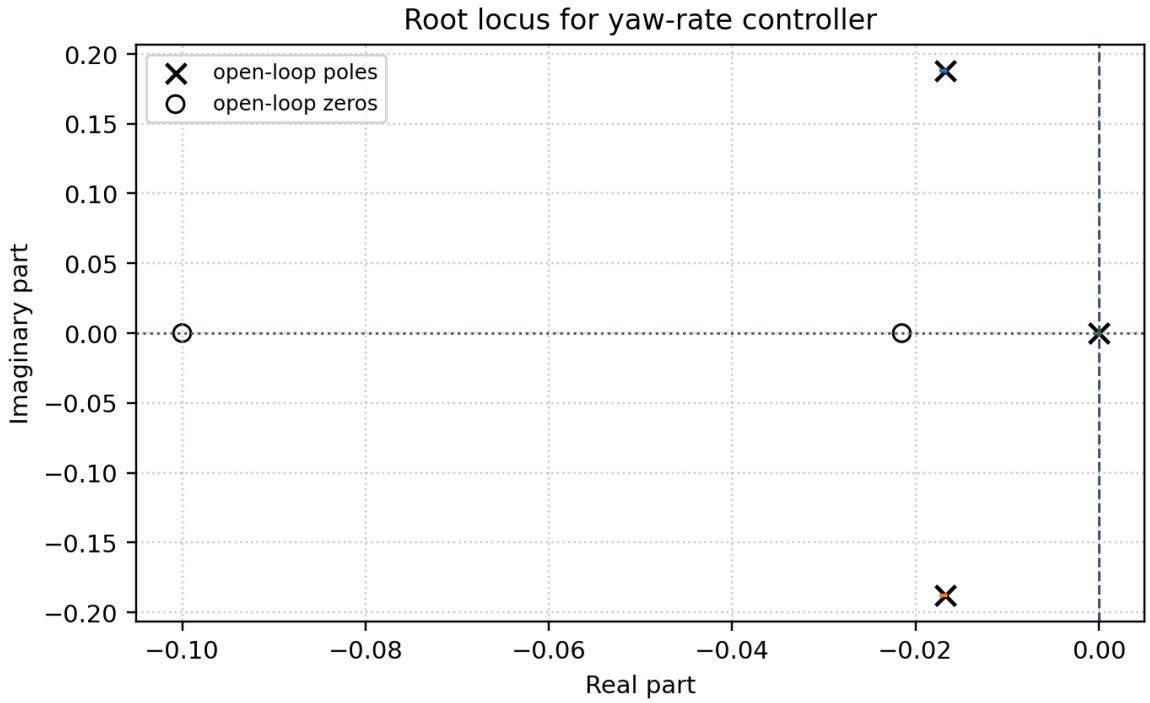


Figure 17. Root locus for yaw-rate controller design at low friction.

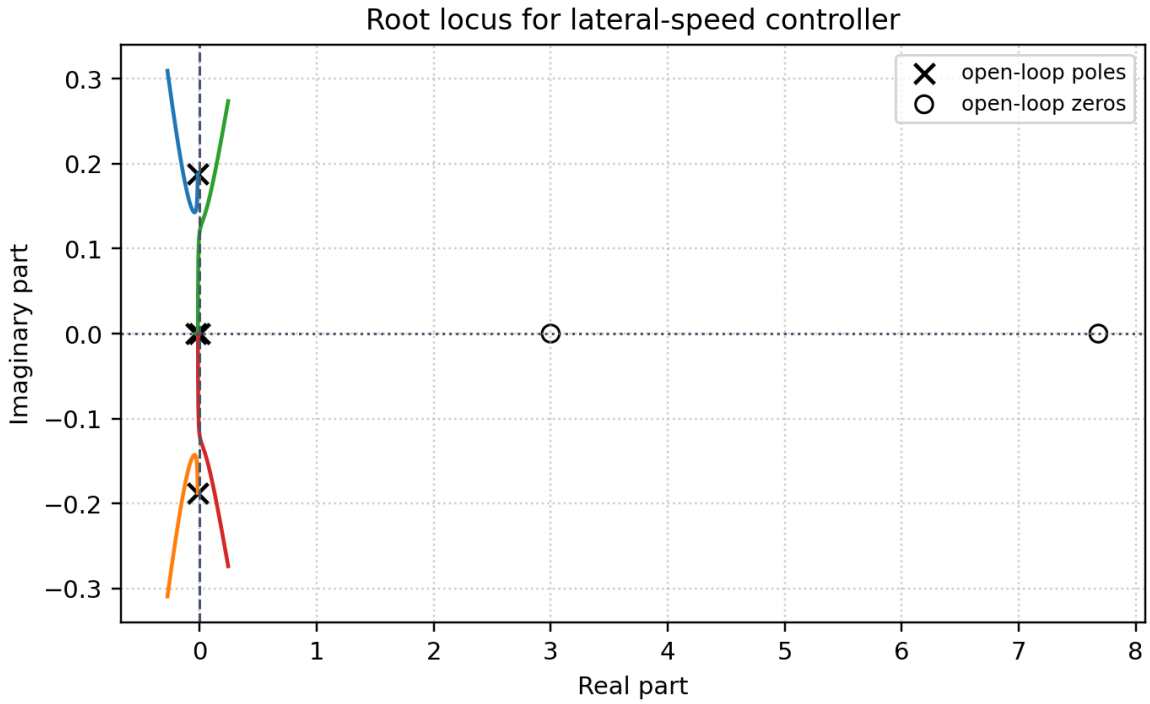


Figure 18. Root locus for lateral-speed controller design at low friction.

Conclusion

The BMW 730 model is stable in forward motion and unstable in backward motion. The yaw-rate output remains observable over the examined forward-speed range, while reachability becomes weak near a low-speed singular condition. The zeros of the transfer functions show that yaw rate is the more suitable measured output for active steering design. The root-locus controllers improve damping in the low-friction case and keep the closed-loop poles in the stable half-plane.